

VxWorks allows control of a computer with split-second accuracy

Wind River, a wholly owned subsidiary of Intel Corporation, is a major player in embedded and mobile software. It has been pioneering computing inside embedded devices since 1981, and its technology is used in more than one billion products. Its flagship product VxWorks was recently used within the Mars rover 'Curiosity.'

Mike Deliman, senior technologist at Wind River, spoke to Ashwin Gopinath of EFY about designing real-time operating system (RTOS) for outer space and the famously complex EDL sequence



MIKE DELIMAN
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What is brewing in the RTOS field? Do you see any particular trend dominating in the near future?

The RTOS field is constantly evolving. Advanced robotics, telepresence and autonomous control all require enormous amount of high-powered deterministic computing. As the need to return more information than raw data grows, these systems will need to do more of number crunching/application processing than had been done by ground and control systems back at the lab. This, in turn, means remotely deployed systems will need more kinds of applications running on them simultaneously to achieve their objectives.

We are going to see an increased need for multi-core support and virtualisation, increased need for robust fault-tolerant supervisory systems, and increased support for Open standards and systems.

How did VxWorks help the Mars rover Curiosity to tackle the so-called '7 minutes of terror' in the entry, descent and landing sequence (ED&L)?

Wind River VxWorks RTOS serves as the OS inside flight computers on Mars rover Curiosity. Just as a Windows or Linux operating system makes your desktop or laptop computer (or cell phone) useful to you as a device, the OS inside flight computers is what makes them a functional space robot. The OS allows programmers to make the computer

do all the things that their applications need, fulfilling the purpose of their device.

In this case, applications included for ED&L procedure take readings from various avionics packages and trigger rockets and other devices to control the landing process. The OS controls the entire process with precision and determinism.

VxWorks was actually in control of the spacecraft's flight from the minute the rocket took off. The craft is also able to upgrade software packages while in-flight. For instance, the craft did not have instructions for ED&L until perhaps a week before it needed them. Once it had landed safely, the instructions that tell it how to be a robotic geologist were sent up.

How was the RTOS of Curiosity customised for this particular mission?

One of the amazing things about VxWorks is its flexibility. It actually did not need to be customised much beyond the layers of software called board support package (BSP) that let VxWorks run on space computers. Even the BSP follows a standard format, which makes it easier to use VxWorks to create systems.

From this point of view, VxWorks operating system inside Curiosity is the same as in more than a billion devices running on the Earth every day. This flexibility is part of what allows VxWorks to be used inside devices like Curiosity.